Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
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EX6 – Man in the Middle Attack in Diffie Hellman Key Exchange

Code:

Alice (Server):

```
import socket

import random

import sympy

import json


P = sympy.randprime(10000, 50000)

G = sympy.primitive_root(P)

X_A = random.randint(2, P - 2)

Y_A = pow(G, X_A, P)


server = socket.socket()

server.bind(("127.0.0.1", 10000))

server.listen(1)

print("Waiting for Bob (but will recieve from attacker instead)...")

conn, addr = server.accept()

payload1 = json.dumps({

"P": P,

"G": G,

"Y_A": Y_A

})

conn.sendall(payload1.encode('utf-8'))
```

```
payload2 = json.loads(conn.recv(1024).decode('utf-8'))
KA = pow(int(payload2["Y_D1"]), X_A, P)

print(f"The prime number is {P}")
print(f"The primitive root of the prime number is {G}")
print(f"Randomly generated private key is {X_A}")
print(f"Computed public key is {Y_A}")
print(f"Public key from attacker is {payload2["Y_D1"]}")
print(f"Common key between Attacker and Alice that Alice has is {KA}")

conn.close()
```

Bob (Server):

```
import socket
import random
import json

server = socket.socket()
server.bind(("127.0.0.1", 9000))
server.listen(1)
print("Waiting for Alice (but will recieve from attacker instead)...")
conn, addr = server.accept()
payload = json.loads(conn.recv(1024).decode('utf-8'))

P = payload["P"]
G = payload["G"]
X_B = random.randint(2, P - 2)
Y_B = pow(G, X_B, P)
KB = pow(int(payload["Y_D2"]), X_B, P)

print(f"The prime number is {P}")
```

```
print(f"The primitive root of the prime number is {G}")
print(f"Randomly generated private key is {X_B}")
print(f"Computed public key is {Y_B}")
print(f"Public key from attacker is {payload["Y_D2"]}")
print(f"Common key between Attacker and Bob that Bob has is {KB}")

conn.sendall(json.dumps({"Y_B": Y_B}).encode('utf-8'))
conn.close()
server.close()
```

Attacker (Client):

```
import socket
import random
import json

sa = socket.socket()
print("Attacker sends Y_D1 to Alice pretending to be Bob")
sa.connect(("127.0.0.1", 10000))
payload1 = json.loads(sa.recv(1024).decode('utf-8'))

P = payload1["P"]
G = payload1["G"]
Y_A = payload1["Y_A"]
X_D1 = random.randint(2, P - 2)
X_D2 = random.randint(2, P - 2)
Y_D1 = pow(G, X_D1, P)
Y_D2 = pow(G, X_D2, P)
sa.sendall(json.dumps({"Y_D1": Y_D1}).encode('utf-8'))

sb = socket.socket()
print("Attacker sends Y_D2 to Bob pretending to be Alice")
sb.connect(("127.0.0.1", 9000))
```

```
payload2 = json.dumps({
"P": P,
"G": G,
"Y_D2": Y_D2
})
sb.sendall(payload2.encode('utf-8'))

payload3 = json.loads(sb.recv(1024).decode('utf-8'))
KAA = pow(int(payload1["Y_A"]), X_D1, P)
KAB = pow(int(payload3["Y_B"]), X_D2, P)

print(f"The prime number is {P}")
print(f"The primitive root of the prime number is {G}")
print(f"Randomly generated private key for Alice is {X_D1}")
print(f"Randomly generated private key for Alice is {X_D2}")
print(f"Computed public key for Alice is {Y_D1}")
print(f"Computed public key for Bob is {Y_D2}")
print(f"Common key between Attacker and Alice that Attacker has is {KAA}")
print(f"Common key between Attacker and Bob that Attacker has is {KAB}")

sa.close()
sb.close()
```

Output:

Alice (Server):

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\MITM Attack Diffie Hellman> python .\alice.py
Waiting for Bob (but will recieve from attacker instead)...
The prime number is 15749
The primitive root of the prime number is 2
Randomly generated private key is 12780
Computed public key is 15473
Public key from attacker is 5760
Common key between Attacker and Alice that Alice has is 9313
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\MITM Attack Diffie Hellman>
```

Bob (Server):

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\MITM Attack Diffie Hellman> python .\bob.py
Waiting for Alice (but will recieve from attacker instead)...
The prime number is 15749
The primitive root of the prime number is 2
Randomly generated private key is 7955
Computed public key is 7004
Public key from attacker is 10653
Common key between Attacker and Bob that Bob has is 6132
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\MITM Attack Diffie Hellman>
```

Attacker (Client):

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\MITM Attack Diffie Hellman> python .\attacker.py
Attacker sends Y_D1 to Alice pretending to be Bob
Attacker sends Y_D2 to Bob pretending to be Alice
The prime number is 15749
The primitive root of the prime number is 2
Randomly generated private key for Alice is 10947
Randomly generated private key for Alice is 2180
Computed public key for Alice is 5760
Computed public key for Bob is 10653
Common key between Attacker and Alice that Attacker has is 9313
Common key between Attacker and Bob that Attacker has is 6132
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\MITM Attack Diffie Hellman>
```